

HAT-P-54b

R-1133

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_241120 · created 2026-07-30T23:18:02+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460634.932 +/- 0.0053 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.094 +/- 0.045
Transit depth (Rp/Rs)^2	0.88 +/- 0.85 [%]
Orbital Inclination (inc)	82.3 +/- 2.38
Airmass coefficient 1 (a1)	0.358 +/- 0.012
Airmass coefficient 2 (a2)	1.0 +/- 0.21
Scatter in the residuals of the lightcurve fit is	3.35 %
Best Comparison Star	None
Optimal Aperture	2.7
Optimal Annulus	12.96
Transit Duration (day)	0.021 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.094 ± 0.045 vs 0.1572 (deviation 1.12σ)
Duration fit vs published	0.021 d vs 0.075 d (deviation 1.8σ)
Mid-time O–C	-7.3 min
Depth SNR	1.7
Residual scatter	3.35 %

Light curve

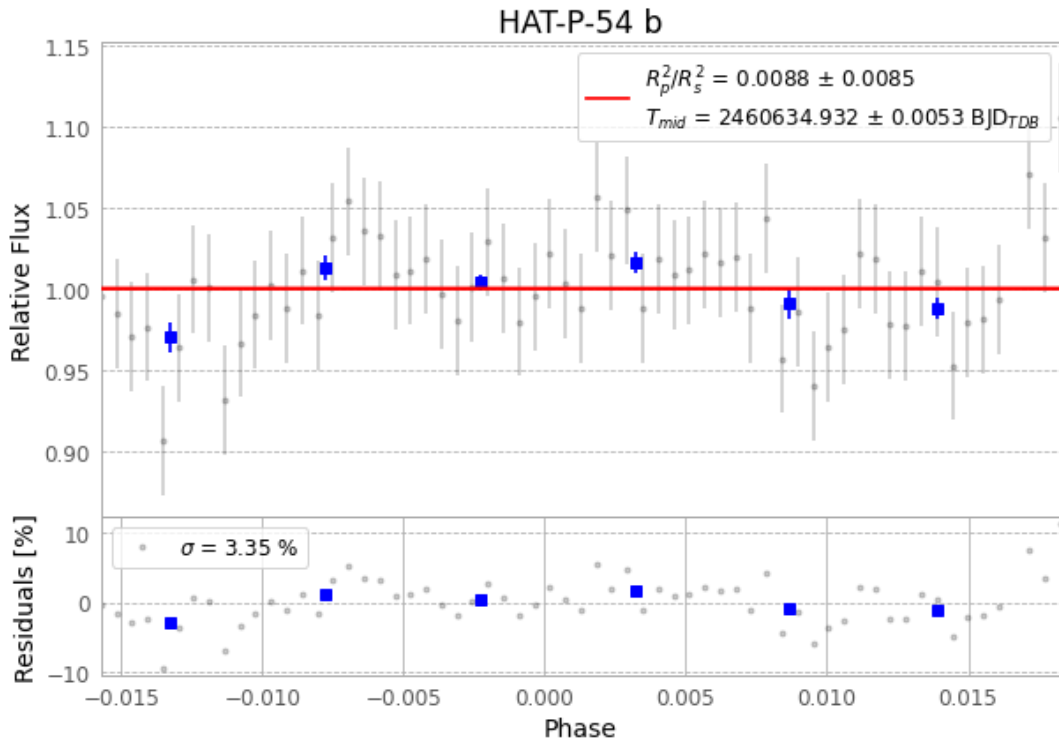


Figure 1 — FinalLightCurve_HAT-P-54 b_2024-11-20.png (SHA-256 156936919c2a4068...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [415, 264], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7dd7bdc37984c3bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

63 FITS frames (41 MB), HATP-54241120084815.FITS → HATP-54241120115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-241120.log (b012af91ba82...), FinalLightCurve_HAT-P-54 b_2024-11-20.png (156936919c2a...), FinalLightCurve_HAT-P-54 b_2024-11-20.csv (26e2d7868e01...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:18Z.